The Trade-off Was in the Labels: Causal Supervision for Turn-Aware Streaming ASR

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Abstract

Every voice agent must decide, in real time, whether the user has finished talking. The deployed default decides from acoustics alone — a voice-activity detector plus a silence timeout — and frontier commercial recognizers have instead moved the decision into the ASR model, where the words are. Every member of this *turn-aware* class is closed, and no training recipe for the capability has been published. We trained one in the open, and first hit what looked like a fundamental trade-off: across eight supervision recipes, every checkpoint either fired eagerly mid-utterance or barely fired at all, and training oscillated between the two behaviors. A one-variable intervention exposed the trade-off as an artifact: appending one second of silence to the evaluation clips raised a "broken" model's fire recall from 0.10 to 1.00. The labels were the problem. Supervision inherited from offline corpora defines end-of-turn using audio *after* the decision point — information a streaming model can never have. One causal labeling rule, with nothing else changed, gives monotone training and a single checkpoint that dominates all eight predecessors: 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute, an operating point no silence timeout reaches at any setting. The same leak then recurred on a second axis: a context prefix supplied to bias transcription became a copyable shortcut, and a wrong user profile was written into the transcript 40% of the time; training examples in which context and audio deliberately disagree drive that intrusion to zero while keeping a +28.9 pp entity-recall benefit. The result is, to our knowledge, the first open turn-aware streaming ASR system with a published training recipe: one small model that transcribes, detects end-of-turn semantically, handles dictation, and grounds transcription in session context. Weights, recipe, and benchmark are released.

Code: `https://github.com/19PINE-AI/turn-aware-asr`
Website: `https://01.me/research/turn-aware-asr`

1 Introduction

A voice agent that cannot tell when you have finished speaking is unusable in two symmetric ways: it interrupts you mid-thought, or it stares at you after you stop. The industry default delegates this decision to a cascade — a voice-activity detector (VAD) plus a fixed silence timeout, feeding a separate recognizer — and the cascade's problem is structural. Humans exchange turns with median gaps around 200 ms [Stivers et al., 2009, Levinson and Torreira, 2015], deployed timeouts sit at 500–1000 ms [Skantze, 2021], and pauses *inside* turns routinely exceed pauses between turns, so no threshold is both fast and safe (§2). The information that resolves the ambiguity — is this thought complete? — is in the words, and the words live inside the recognizer. Frontier commercial systems have drawn the conclusion and moved end-of-turn into the ASR model itself: Deepgram's Flux emits end-of-turn events from the recognizer's forward pass [Deepgram, 2026], and OpenAI's Realtime API waits longer after a trailing "ummm" than after a completed sentence [OpenAI, 2025]. We call this class **turn-aware streaming ASR**: one model that transcribes while deciding, from semantics and silence jointly, whether the turn is over. Every commercial member of the class is closed. The open landscape offers pieces — open weights with no training account (Kyutai STT, Parakeet-EOU), open turn *detectors* that do not transcribe (Smart Turn, LiveKit, TEN) — but no recipe for the combination (§10).

This paper reports what happened when we tried to train the capability in the open, because the interesting part is not that it worked; it is why it kept failing. Eight successive supervision recipes, all on the same small open model (Qwen3-ASR-0.6B, Shi et al., 2026) with the same adapter and data volume, produced the same dichotomy: checkpoints that fired promptly at turn ends also fired spuriously mid-utterance, checkpoints that never fired spuriously would not fire at all, and later recipes oscillated between the two behaviors for entire training runs. The record looked like a fundamental recall-versus-precision frontier, and the project postmortem at recipe eight recommended shipping two checkpoints, one eager and one conservative (§3).

The frontier was an artifact, and one experiment exposed it. Appending 1.0 s of silence to the offline evaluation clips — restoring the future that a live microphone always delivers — took the "broken" eighth recipe's fire recall from 0.10 to 1.00 (§4). The diagnosis is a form of target leakage, transposed into time: offline corpora are cut at exactly the boundaries a streaming model must detect, and offline transcripts record who spoke next, so supervision built naively on either defines the label using audio *after* the decision point. A streaming model can never see that audio. Labels built on it do not merely add noise; they partition the training data into internally consistent pools with incompatible optima, which is what manufactures both the oscillation and the apparent trade-off. We call such labels *clairvoyant*. The fix is one causal labeling rule — fire iff the words so far are semantically complete *and* at least 0.3 s of silence has been observed — plus a minimal-pair construction that forces the model to key on observables (§5). With nothing else changed, training becomes monotone and a single checkpoint dominates all eight predecessors: on a deployment-matched streaming benchmark it reaches 0.97 boundary recall at 0.39 s median latency with 0.3 false fires per speech-minute, an operating point outside the entire silence-timeout family (Figure 1; §6).

We then hit the same failure a second time, in a different place. The model takes a text context prefix (a user profile, a hotword list) that biases transcription for the whole session, lifting entity recall by +28.9 pp on entity-dense earnings-call audio. Trained only on examples where that prefix matched the audio, the model learned to *copy* the prefix rather than listen: a wrong-user profile wrote the wrong name into the transcript 40% of the time. The leak has the same shape — the training target depended on a signal the model cannot trust at decision time — and the same repair: counterfactual examples in which context and audio deliberately disagree and the target follows the audio, which drive intrusion to zero without losing the biasing benefit (§7). Two independent instances of one failure class, each diagnosed by intervention and cured by the same construction, are the paper's central evidence that the principle is general rather than an anecdote about endpointing.

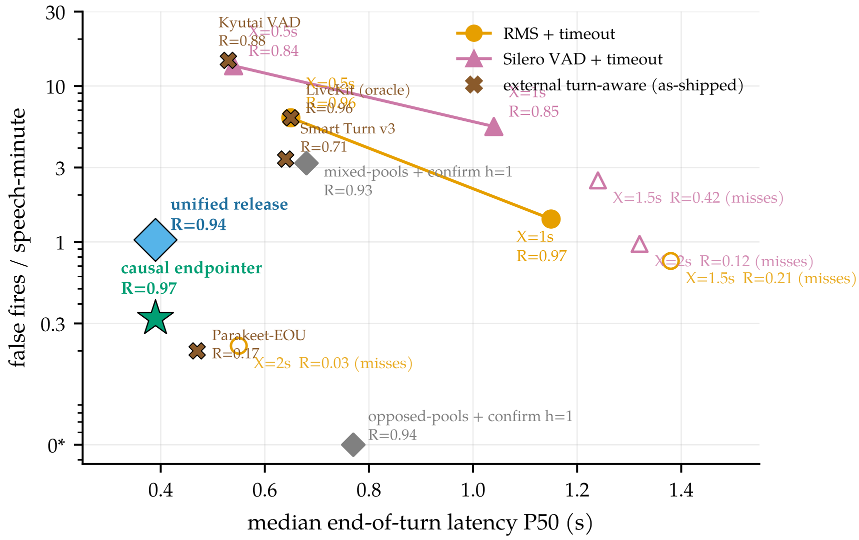

Figure 1. The causal model escapes the timeout curve. Median end-of-turn latency vs. false fires (log scale; 0* plotted at 0.05) on identical audio, detector, and scoring (§5.2). A silence timeout can only slide along its curve; the causally supervised model sits strictly inside the whole family because it reads completeness, firing 0.39 s after a finished thought while holding through a 1.4 s mid-sentence pause.

Our contributions:

1. **A diagnosis with an intervention** (§3–§4): clairvoyant labels — streaming supervision that depends on post-decision input — manufacture training oscillation and phantom capability trade-offs; a one-variable experiment falsifies the frontier that eight recipes of data engineering had been fighting.

2. **A causal training recipe, and the first open turn-aware streaming ASR system** (§5–§6): one labeling rule and a minimal-pair construction yield a single small model (LoRA on Qwen3-ASR-0.6B, ∼15k synthesized examples, hours on one GPU) that transcribes, emits in-transcript end-of-turn events, and beats every silence-timeout setting; weights, recipe, and a deployment-matched benchmark — under which our historical model ranking *inverts* — are released.

3. **A second instance, and the general principle** (§7): the contextual analog of the temporal leak (context copying; 40% wrong-profile intrusion) is cured by the analogous counterfactual construction (intrusion to zero, biasing intact), folding conversation, dictation, and context grounding into one checkpoint; we distill both instances into a checklist for any streaming decision trained from offline logs (§9).

2 The task, and why a threshold cannot solve it

Two turns no timeout can get right

1. The pause that means "wait." *"My number is four one five . . . (0.9 s) . . . five five five . . . (1.1 s) . . . zero one nine two."* A 0.7 s timeout fires twice mid-number. The silence carries no evidence that the turn is over, but the words do: seven digits of a ten-digit pattern predict continuation. Production teams report exactly this failure [LiveKit, 2026].

2. The completion that means "go." *"Hello? I'm still here."* The thought is complete the instant the last word ends; a human would respond within $\sim$200 ms [Stivers et al., 2009]. A 1.0 s timeout makes the user wait five times that long, for nothing.

The two examples fail in opposite directions: the first needs the system to hold *through* long silence, the second to fire at near-zero silence. No single threshold can do both, because the gap between the cases is not measurable in seconds of quiet. On our own conversational benchmark the silence-gap distributions of within-turn pauses and between-turn boundaries overlap almost completely, and the pauses are the *longer* of the two (§4); Skantze [2021] documents the same overlap across the dialogue literature, and the resulting no-good-threshold dilemma has defined cascade endpointing since Raux and Eskenazi [2008]. The resolving signal is semantic completeness, which is exactly what a recognizer computes on the way to a transcript. §7 builds probes for these two scenarios and closes them with training data alone.

Transcription has its own version of the argument. A voice agent hears names, tickers, account emails — tokens nearly unrecoverable from acoustics alone but obvious given who is calling about what. Every major commercial STT ships a context mechanism, yet none publishes controlled uplift numbers, and the open model we build on describes its context facility in one qualitative sentence [Shi et al., 2026]. A turn-aware model with a context slot can ground the whole session's transcription in the profile the agent already has; we measure exactly this in §7.

The system under study. A single-channel audio stream arrives in 0.5 s chunks. At each chunk boundary the system must extend a running transcript and decide whether the turn has ended — an irreversible, latency-sensitive event. A text context prefix may be supplied at session start. We start from Qwen3-ASR-0.6B [Shi et al., 2026], an open unified speech model (AuT audio encoder + dense Qwen3 decoder) with strong offline WER but no endpointing, claim two reserved vocabulary slots as marker tokens <EAGER_END_SPEECH> and <END_SPEECH>, and LoRA-fine-tune [Hu et al., 2022] the decoder (r=16, α=32, q/k/v/o projections, plus the two marker embedding rows) to emit them inside the transcript at endpoint moments. Training data is synthesized from LibriSpeech [Panayotov et al., 2015] and AMI [Carletta et al., 2005]; training takes hours on one GPU. At inference, three interpretable knobs sit behind the model (Figure 2): an RMS **energy gate** that never *starts* a segment on a silent chunk, a **confirm horizon** h that holds a fire candidate for h further silent chunks before signaling END, and a **max-segment force-flush** that bounds the committed transcript segment. Every reported arm states its knob settings.

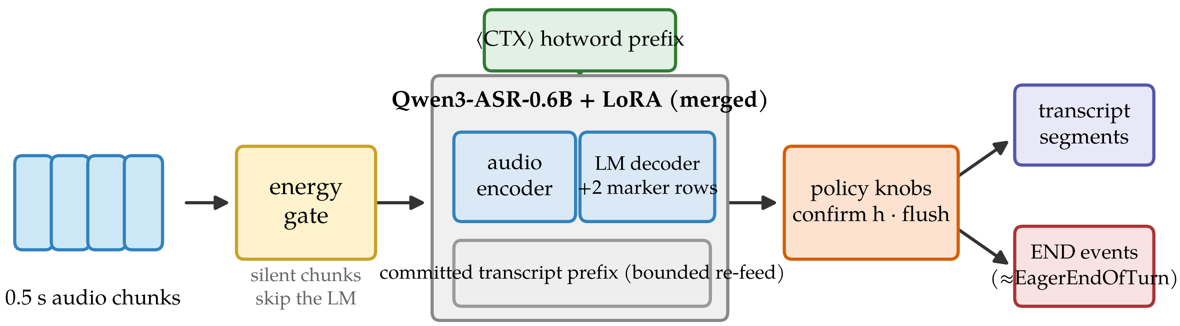

Figure 2. System shape. Audio chunks pass an energy gate; the merged LoRA model transcribes the current bounded segment with the `<CTX>` prefix in the system slot and may emit marker tokens; two policy knobs sit behind the model. Outputs are transcript segments plus END events, the same event shape as Flux's `EagerEndOfTurn`.

Table 1. The landscape (verified July 2026; discussion in §10). "Turn detectors" are separate models beside a recognizer; "cascade" is VAD + silence timeout + separate ASR, the deployed default and our measured baseline.

	This work	Flux / `sem.vad`	Kyutai STT	Parakeet-EOU	Turn det.	Cascade
Streaming ASR	✓	✓	✓	✓	—	✓
In-model end-of-turn	semantic	semantic	sem. head	token	sep. model	timeout
Open weights	✓	—	✓	✓	mostly	✓
Turn-training recipe	✓	—	—	—	Smart Turn	n/a
Context biasing (numbers)	✓ (+28.9 pp)	—	—	—	—	ASR-dep.

3 Eight recipes hit the same wall

The first eight supervision recipes shared everything — architecture, adapter, data volume, source corpora — except the labels, and their record reads as a dose–response study of what contradictory supervision does to a model. Table 2 compresses it. The first recipe supervised on offline clips alone: fire at every clip end. It scored perfectly offline and was unusable live, firing 89 times per speech-minute mid-utterance. Successive recipes added pools intended to fix this — truncated-speech holds, trailing-silence fires, hold pools on disjoint audio — and every addition moved the model *along* an apparent recall-versus-precision frontier, never off it. Two behaviors recurred across the family. Checkpoints were bimodal: eager ones fired mid-utterance, conservative ones missed turns, and no checkpoint did both jobs. And late recipes did not converge at all: holdout accuracy oscillated for the entire run between a *fire-mode* attractor (fire class ≈1.0, hold class ≈0.4) and a *no-fire-mode* attractor (fire ≈0.1, hold ≈1.0), as in Figure 3 (left).

By the eighth recipe the working conclusion was that the two behaviors lay on a fundamental Pareto frontier, and the postmortem recommended shipping two checkpoints, one eager and one conservative. That conclusion was wrong.

Table 2. The supervision-composition record. Eight recipes, one architecture; each row adds one labeled pool (cumulative). "Early" latencies are negative: the model fires seconds before the turn ends. Row 7 anticipates §5.

#	Supervision change (cumulative)	Observed behavior
1	offline endpoint pool only: fire at every clip end	fires eagerly and "accurately" offline (100% fire recall); in streaming replay, fires 89× per speech-minute mid-utterance
2	+ truncated-speech hold pool (11%→38% of mix)	diminishing returns: median fire timing improves only $-13.1 \to -9.6 \to -7.6$ s (still seconds early) as the dose rises
3	+ trailing-silence fire pool	streaming latency crosses zero (P50 +0.32 s) but offline fire recall regresses 100→82%; the "trade-off" is first declared
4	+ complete utterances with *and* without silence tails, both labeled fire	silence becomes optional again; a strictly intermediate point on the same frontier; frontier declared fundamental
5	+ hold pool on the *same* audio as existing fire examples	collapse: opposite labels on identical inputs; the model stops firing on meeting audio entirely (fire recall 0%)
6	+ hold pool on *disjoint but identically distributed* audio	bimodal oscillation between fire-mode and no-fire-mode attractors for the whole run (Figure 3, left); no checkpoint dominates; postmortem recommends shipping two checkpoints
7	**causal relabel (§5):** fire iff complete *and* ≥0.3 s observed silence	monotone training, both classes at ceiling simultaneously; single checkpoint dominates every predecessor (§6)

4 One second of silence: the intervention, and the diagnosis

The experiment. The frontier's fire-recall axis died by a one-variable intervention on the *evaluation*: we appended 1.0 s of silence to each clip of the legacy offline benchmark and re-scored the historical checkpoints (Table 3). The "broken" eighth recipe's fire recall rose from 0.10 to 1.00 and the fifth's from 0.82 to 0.98, while the offline-labeled first recipe — which never depended on hearing silence — was the unchanged control. Under deployment-realistic input, all three fire essentially perfectly on complete utterances. The capability axis that four recipes of data engineering had been fighting was an artifact of where the evaluation clips ended.

Table 3. The silence-append intervention. Fire recall on complete single utterances from the legacy offline benchmark, before and after appending 1.0 s of silence to each clip.

Model (supervision recipe)	original clips	+1.0 s silence	Δ
offline-labeled (rows 1–2)	1.00	1.00	0.00
mixed-pools (row 3)	0.82	0.98	+0.16
opposed-pools (row 6)	0.10	1.00	+0.90

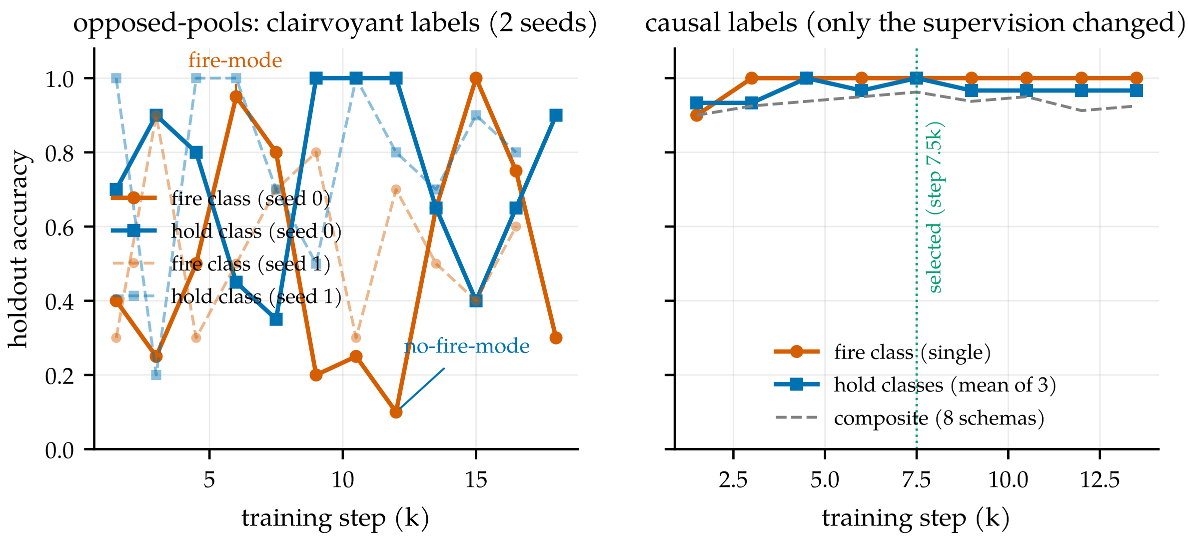

Figure 3. The fingerprint of contradictory supervision. Left: under the opposed-pools recipe, holdout accuracy oscillates for the entire run between a fire-mode and a no-fire-mode attractor, reproducibly across two training seeds (solid, dashed) — the oscillation is a property of the labels, not a seed. Right: the causal recipe of §5 — same architecture, adapter, and data volume, only the labels changed — climbs monotonically, with both classes at ceiling simultaneously from step 3k on.

The diagnosis. Why did the clips end where they did? Offline ASR datasets are segmented by forced alignment, so clips are cut at the end of speech, and evaluation clips inherit the cut. An offline endpoint benchmark built on such clips therefore demands a fire on audio that ends *during or immediately after* speech — a condition deployment never presents, because on a live channel silence keeps arriving after a real speaker stops. The streaming pools taught the opposite, correctly: never fire before silence is observed. The condition "complete utterance, no trailing silence yet" thus carried the label *fire* in one pool and *hold* in the other (Figure 4a) — in the eighth recipe, 50/50 label noise on the exact decision the model exists to make. A second mechanism had the same structure: the historical pools separated "disfluent pause, hold" from "turn boundary, fire" using the transcript's segmentation — same speaker continues, disfluency; another speaker follows, boundary. Measured at the decision point, mid-pause, the two classes are statistically indistinguishable: their silence-gap distributions overlap almost completely (medians 1.39 s vs. 0.83 s, and the disfluent pauses are the longer ones; Figure 4b). The label is a function of who spoke *next* — of the future.

Both mechanisms are the same error, and nothing about it is specific to speech. It is target leakage transposed into time: the label encodes information from beyond the decision point, so the model is being graded on knowledge it cannot have.

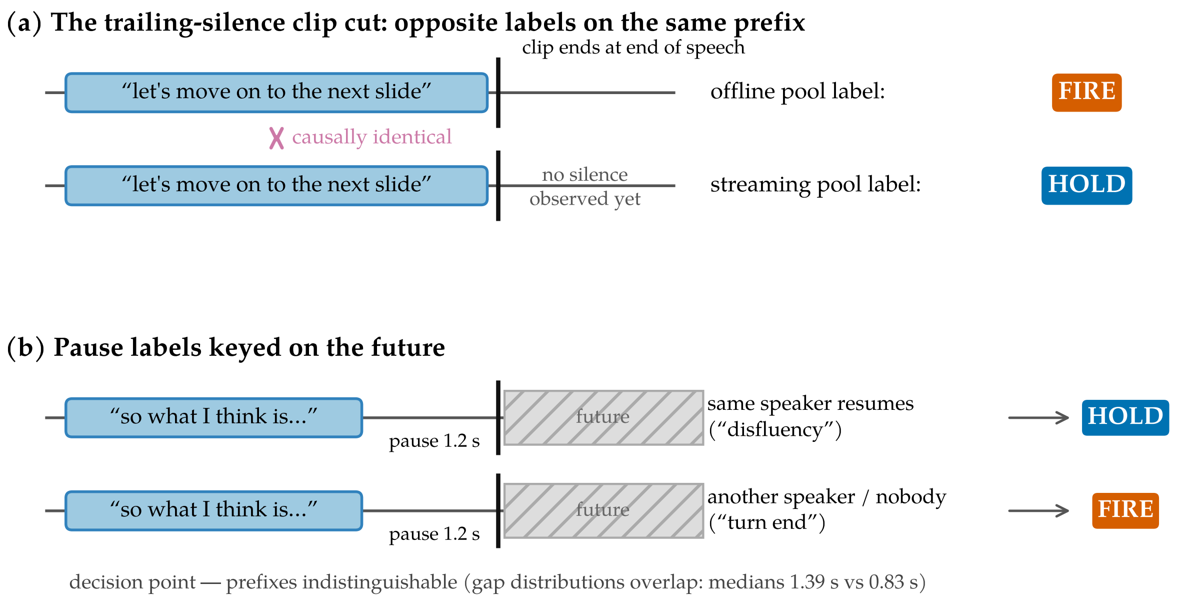

Figure 4. Two ways offline-derived supervision leaks the future. (a) Offline clips inherit forced-alignment cuts at end of speech, so the offline pool demands a fire on exactly the prefix the streaming pool labels *hold*. (b) Disfluency-vs-boundary pause labels are keyed on who speaks next; at the decision point the two classes' silence-gap distributions overlap almost completely.

Clairvoyant labels

A label for a streaming decision at time t is *clairvoyant* if its value depends on input after t. Pools containing clairvoyant labels do not merely add noise: because the dependence is systematic, they partition the data into internally consistent subsets with incompatible optima. The observable symptoms are (a) training-time oscillation between mode attractors and (b) apparent capability trade-offs across checkpoints — a phantom Pareto frontier. **Principle: every training label and every evaluation target for a streaming decision must be computable from the input up to the decision point.**

Why oscillation, specifically. Rows 5 and 6 of Table 2 form a natural two-condition experiment on how a contradiction expresses itself. When opposite labels sit on literally identical audio (row 5), the conflict is visible to the loss on every example, and the model settles deterministically on one side: it simply stops firing. When opposite labels sit on disjoint but identically distributed audio (row 6), no single example exposes the conflict; each pool is separately learnable, incompatible only in aggregate, and training circulates between mode attractors that each satisfy one pool. The oscillation is not instability to be scheduled away. It is the loss surface reporting a contradiction that no individual example states. A logit-level check locates the circulation in the weights rather than at a decision threshold: the model's marker-token probability (marginalized over the deterministic two-token marker path) swings between snapshots in lockstep with the argmax (r=0.98 on the contested fire schema), yet not one held-out example sits persistently in the ambiguous $[0.2, 0.8]$ band across the run and several reverse from confidently-held to confidently-fired — the signature of a threshold

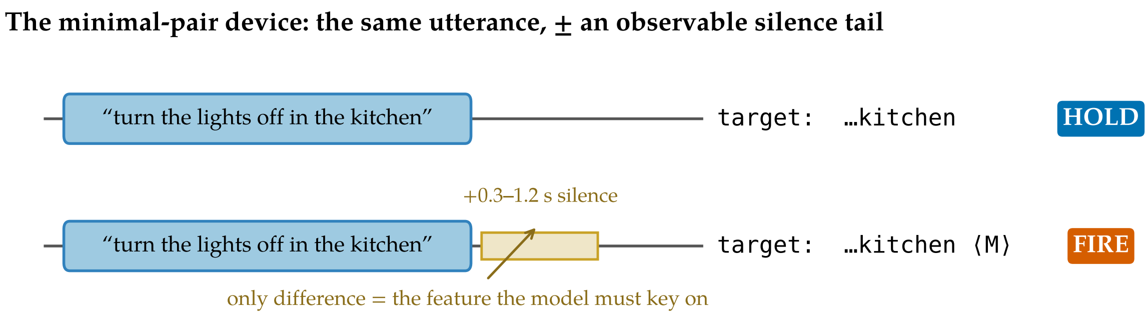

Figure 5. The minimal-pair device. Each complete utterance appears twice: without a silence tail (target: transcript only) and with a 0.3–1.2 s tail (target: transcript + marker). The only difference between the opposite-labeled examples is the observable the model must key on.

flapping at a stable probability is absent.

5 The fix: causal labels, and a benchmark to match

5.1 One labeling rule

The entire turn capability comes from the labels. One rule generates every training example: *emit <EAGER_END_SPEECH><END_SPEECH> at a point iff the speech so far is semantically complete AND* $\geq$*0.3 s of silence has elapsed since the last speech.* Both conditions are functions of the audio prefix. Completeness is decided by a transcript heuristic (trailing filler or connective, mid-word cut, or very short non-acknowledgment $\Rightarrow$ incomplete; Appendix A); silence is decided by the synthesized waveform itself.

Two constructions carry the semantics-versus-silence discrimination that §2 asked for (Table 4). The **minimal pair** (schemas 1–2, Figure 5) presents the same complete utterance twice, with and without an observable silence tail, with opposite targets: completeness alone must not fire; the model has to *see* the silence. This is the direct antidote to the row-4 pathology of Table 2, where silence had been made optional. The **pause pair** (schemas 4–5) presents two-utterance sequences whose gap is bridged or fired depending only on whether the words before the pause are complete: "…and then, *(pause)*" holds; "…that's everything. *(pause)*" fires, even if the speaker later resumes. Quick resumes after a legitimate fire become two transcript segments — a measured cost, not an error (§5.2) — because at decision time the correct answer does not exist on the channel. Schemas 7–8 cover the silence-heavy channel shape that pure-utterance corpora never show. Pair examples use both same- and different-speaker sources: the fire decision keys on completeness plus silence, never on speaker identity, which single-channel deployment cannot observe. Removing each construction in turn (Appendix C, Table 10) isolates what it buys: without the minimal pair, premature firing returns (false fires 1.08→5.28 per speech-minute); without the silence schemas, the model sprays markers on silence (8→1458 ungated); the pause pair is partly redundant with the complete-A gap schema.

5.2 A deployment-matched benchmark

If clairvoyant labels are the disease, evaluation is where it enters: the legacy offline benchmark did not just mis-score the historical models, it drove four recipes of wasted data engineering. We replaced it with one deployment-matched protocol and applied the causality principle to

Table 4. The eight training schemas (∼12k examples; LibriSpeech and AMI roughly 50/50). M = <EAGER_END_SPEECH><END_SPEECH>.

#	Construction	Target
1	complete utterance + 0.3–1.2 s silence tail	`text` M
2	*same utterances as #1*, tail ≤0.1 s	`text`
3	speech truncated mid-utterance at 30–80%	partial `text`
4	complete A + gap 0.3–2.5 s + B (+ tail)	`A` M `B` M / `A` M `B`
5	*incomplete* A + gap + B + tail	`A B` M
6	complete utterance + long silence (2–4 s)	`text` M
7	pure silence 0.5–4 s	(empty)
8	leading silence 0.5–2 s + utterance + tail	`text` M

the benchmark itself.

Material and ground truth. Continuous single-channel stretches from held-out AMI meetings: one target speaker's utterances at their true timeline offsets, silence between them, 30–60 s per stretch — the voice-assistant shape, where audio never ends at a speech boundary. Each utterance-end boundary is classified by what is observable *on this channel*: **turn-final** (same-speaker gap ≥2 s), where the system should fire; **continuation** (gap in $[0.3, 2)$ s, same speaker resumes), where firing is a measured segmentation cost rather than an error, because the correct answer depends on the future; and **internal** (pauses inside utterances, or <0.3 s after speech), where firing is a false fire. Metrics: boundary recall within $[-0.25, +1.5]$ s, false fires per speech-minute, latency percentiles, resume rate, streaming WER of flushed transcripts, and per-chunk compute. An early version of the benchmark itself contained a clairvoyant rule — promoting boundaries to turn-final when another meeting speaker interleaved, which is inaudible on this channel; our own checklist (§9) caught it, and all numbers use the corrected classification (Appendix B).

Two properties of the measurement itself. First, models trained only on speech-initial examples hallucinate on silence: the ungated mixed-pools model sprays markers at ∼1.9 fires per second of silence (r=0.997 against stretch silence content; Figure 6), which makes *ungated* recall uninterpretable — a random 1.9 Hz spammer scores 0.96 against the tolerance window. All comparisons therefore run behind the energy gate, and the causal recipe additionally bakes silence schemas into training. Second, the deployment-matched protocol does not merely re-score the historical models; it re-*ranks* them (Figure 7). The offline champion fires 89 times per speech-minute once audio keeps flowing — its offline strength *is* a premature-fire pathology — while the opposed-pools model, written off at 10% offline recall, turns out to be the best prior endpointer. Both historical ship decisions were wrong, and any team selecting a turn-aware checkpoint on clip-cut offline evaluations should expect the same inversion.

6 Results: one checkpoint dominates

Table 5 reports the relabel. Against its predecessors, the causal model with the gate alone is better on every axis at once: its *raw* false-fire rate beats the mixed-pools model's *confirmed* rate, at half the latency and higher recall. Against the deployed cascade, the comparison is structural.

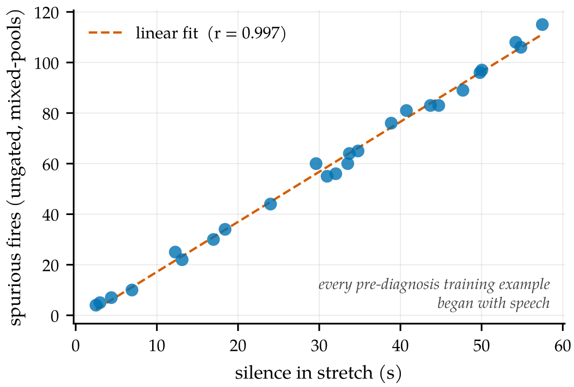

Figure 6. Silence incompetence. Spurious fires of the ungated mixed-pools model track the silence content of each stretch almost perfectly; ungated recall is therefore uninterpretable.

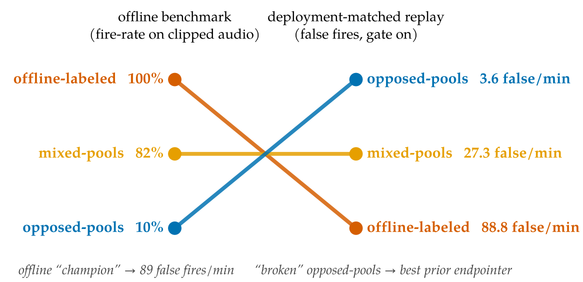

Figure 7. The ranking inversion. Model ranking on the legacy offline benchmark (left) versus the deployment-matched replay benchmark (right). The offline champion is the worst live model.

Table 5. Main endpointing result (deployment-matched replay, energy gate on; the latency-vs-false-fire view is Figure 1). Rows 1–4: the 25-stretch development benchmark (96 turn-final boundaries). Last row: a fresh 2×-larger stretch set drawn after all development ended. †WER mean dominated by long-monologue repetition loops when no force-flush bounds the segment (fresh-set WER *median* 0.30, matching the development set; the §8 force-flush bounds this tail).

Policy	Recall ↑	P50 ↓	P95 ↓	False/min ↓	WER_{mean} ↓
mixed-pools + gate + confirm h=1	0.927	0.68 s	1.27 s	3.2	0.36
opposed-pools + gate + confirm h=1	0.938	0.77 s	1.06 s	0.0	4.96†
causal + gate (ours)	**0.969**	**0.39 s**	**0.70 s**	**0.3**	**1.43**
causal + gate + confirm h=1	0.969	0.89 s	1.20 s	0.0	1.43
causal + gate — fresh 50-stretch set	0.924	0.42 s	0.70 s	0.2	4.63†

A timeout must be long enough to survive within-turn pauses, so it can only trade latency against false fires along one curve; the causal model escapes the curve by reading completeness, combining exactly the pair of behaviors §2 argued no threshold can combine. The full timeout family, swept over both detectors and all settings on both stretch sets (Appendix C): matching the causal model's recall costs a timeout 3× the latency and ∼5× the false fires, and no setting reaches its latency-and-false-fire corner at any recall.

Two further observations. The inference crutches retire: the confirm horizon that earlier recipes needed to be usable cancels only 3 fire candidates for the causal model, versus 241 for the mixed-pools model — the phrase-versus-turn discrimination moved from inference policy into the weights. And the result survives fresh data: because the 25-stretch benchmark was reused during development, we drew a 2×-larger set after all development ended (Table 5, last row). Latency and false fires hold; recall dips 4.5 pp, within the overlap of the two sets' 95% Wilson intervals, and we treat the fresh set as the operating estimate. One caveat by design: quick resumes after a legitimate fire are always segmented (resume rate 1.0), because the correct choice is unknowable at decision time; deployments preferring merged segments pay +0.5 s via the confirm horizon.

External turn-aware systems on the same protocol. We ran the open turn-aware systems of §10 through the identical benchmark — same audio, energy gate, and scoring — as-shipped, sweeping their public thresholds rather than retraining (Figure 1). None reaches the causal model's corner. Smart Turn v3 (an acoustic classifier triggered on VAD silence) sits at 0.71 recall and 3.4 false fires per speech-minute; LiveKit's text detector, handed an *oracle* transcript, reaches 0.96 recall but at 6.25 false fires; NVIDIA's Parakeet-EOU fires so rarely it holds recall to 0.17; Kyutai's semantic-VAD head reaches 0.88 recall only at 14.6 false fires. The detectors are gated on VAD silence by construction and slide along the timeout curve; LiveKit's oracle-transcript number shows the binding constraint is that gating, not transcript quality. To our knowledge this is the first measurement of the open turn-aware class on a common protocol; it is as-shipped, not best-achievable (none was tuned for AMI close-talk).

7 The same leak on a second axis: context

The model's second capability is a text context prefix — user profile, hotword list — that biases recognition for the whole session. Building it produced an independent instance of the §4 failure class, on an axis that has nothing to do with time.

Context biasing works, and survives session length. On Earnings-22 [Del Rio et al., 2022] — spontaneous earnings-call speech dense in tickers, executives, and products, with entities LLM-extracted and filtered to genuine proper nouns — a relevant hotword prefix lifts entity recall from 66.7% to 95.6% (+28.9 pp) while nudging WER down (15.3→15.0%): the prefix acts as a domain hint, not a distortion (Figure 8a). With up to 12 s of unrelated speech interposed between prefix and target words, the advantage stays flat at ≈+30 pp (Figure 8b) — the property a voice agent actually needs, since the user's account entities must help at minute five, not just in the first utterance. The base model's technical report describes this capability in one sentence and publishes no numbers [Shi et al., 2026]; to our knowledge these are the first on this model family, and a concurrent benchmark corroborates the effect across commercial systems [Durmus et al., 2026].

The copying trap. The §2 motivating scenarios get probes: a **dictation probe** that replays ten-digit phone numbers (3-3-4 groups from held-out Free Spoken Digit Dataset speakers, 0.6–1.2 s inter-group pauses) through the exact streaming stack of §5.2, and a **spelled-entity probe** that decodes synthesized utterances spelling uncommon names and email addresses, with and without a user-profile prefix, plus a wrong-user *distractor* profile. The conversational model of §6 fails both, as predicted: it reads a digit group as a finished thought (5.0 premature fires per dictated number, worse than a 0.5 s timeout) and spells emails at 4% exact. We extended the recipe with dictation and context schemas under the same causal discipline — partial phone numbers followed by a pause are labeled *hold* because the pattern predicts continuation; spelled entities are targeted in normalized written form, half with a profile in the context slot — and retrained the same LoRA on the extended pool (∼15k examples).

Dictation was fixed immediately; context introduced the trap. Trained only on examples where the profile matched the audio, the model learned that the context *is* the answer: it copied whatever entity the prefix named, so a wrong-user profile wrote the wrong email into the transcript — 11.3% wrong-profile intrusion, rising to 40% with longer training. This is the §4

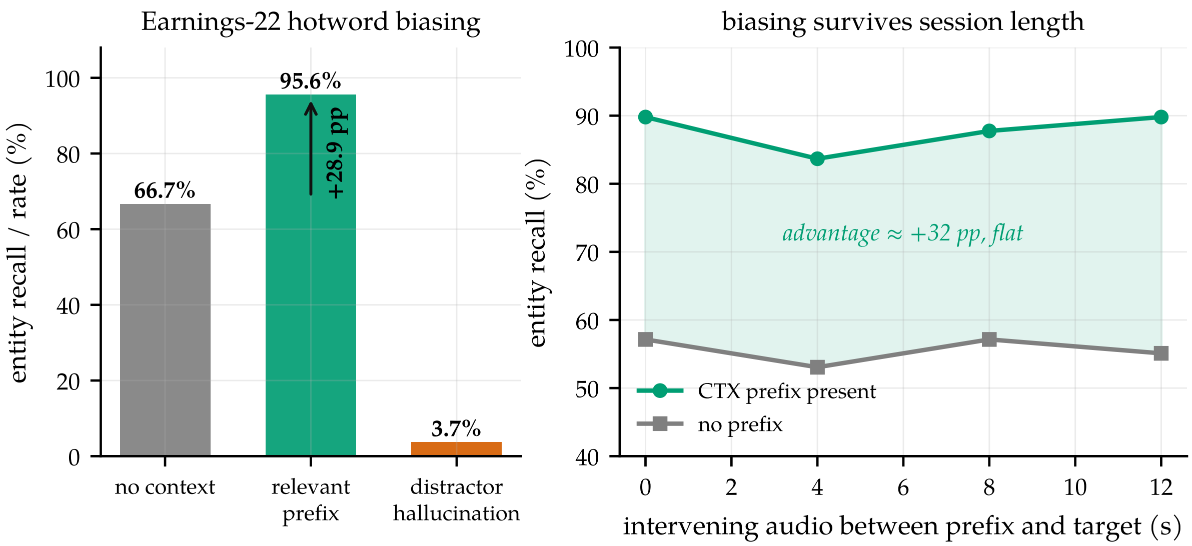

Figure 8. Context biasing, measured. (a) Earnings-22 entity recall without context, with the relevant hotword prefix, and hallucination under a distractor prefix (150 utterances). (b) Recall vs. seconds of unrelated speech between prefix and target: the advantage does not decay with session age.

Table 6. Dictation and spelled-entity probes. Premature fires per dictated 10-digit number, final-boundary recall, digit accuracy; exact-match on spelled names/emails without/with the profile; wrong-profile intrusion. Conversational-only and released rows are scored on enlarged probes (250 sequences; 240 spelled items over 120 identities); the intermediate row uses the smaller original probes. Confidence intervals and a scoring-window sensitivity analysis are in Appendix C.

	Premat./seq ↓	Final rec. ↑	Digit acc ↑	Name −/+	Email −/+	Intrus. ↓
conversational only	4.96	0.82	0.889	0.25 / 0.82	0.04 / 0.58	1.7%
timeout X=0.5 s	2.00	1.00	—	—	—	—
timeout X=1.0 s	0.74	1.00	—	—	—	—
+ dictation (match-ctx)	0.36	0.78	0.996	1.00 / 1.00	0.03 / 0.93	11.3%
+ **distractor (released)**	**0.16**	**0.84**	**0.998**	**0.98 / 1.00**	0.13 / **0.94**	**0.0%**

leak in a new coordinate. There, the target depended on the future, which the model cannot see; here, the target correlated perfectly with a side input the model cannot trust, because at decision time the profile can simply be wrong. The repair is also the same construction. A third of the spelled examples carry a *conflicting* profile — the context names one identity, the audio spells another, and the target follows the audio — exactly as the minimal pair of §5.1 presented the same utterance with and without a silence tail. Counterfactual training drives intrusion to zero at every checkpoint of the run while leaving dictation and matching-context accuracy intact (Table 6): context becomes a prior, not a substitute for listening.

One released model, four behaviors. The released checkpoint endpoints conversation, holds through dictation (0.16 premature fires per number under the shipped scoring window — better than every timeout setting — at 0.998 digit accuracy), grounds spelled entities in context (names 100%, emails 94% exact), and ignores a wrong profile (0.0% intrusion), with no deployment-time checkpoint swap. The shipped dictation window is in fact conservative: a re-analysis scoring each fire by its offset from the true end of speech shows the residual "premature" fires land $\sim$0.15–0.25 s *after* the last digit, just past the window's +0.25 s cutoff; delta-based scoring gives 0.008 genuine mid-sequence fires per number, 0.992 on-time final recall, and zero late fires (Appendix C). On a large held-out replay set (100 stretches, 384 turn-final boundaries) the released model scores 0.938 boundary recall at 1.03 false fires per speech-minute, P50 0.39 s — still strictly inside the timeout family of Figure 1 — and the capability is a property of the recipe rather than a lucky checkpoint: retraining under three seeds gives premature-fire rate 0.17 ± 0.03, email-with-context accuracy 0.90 ± 0.04, and intrusion 0.004 ± 0.006 (Appendix C).

Which frontiers are real. Folding dictation and context into the same rank-16 adapter costs some endpointing precision: 1.03 false fires per speech-minute versus 0.3 for the pure-endpointing checkpoint of §6, a genuine capacity trade-off that we report and do not tune away (a deployment that only needs turn-taking can run the pure checkpoint). The endpoint recall-versus-precision frontier dissolved when the labels became causal; the breadth-versus-precision cost at fixed adapter capacity does not — a distinction the §4 principle exists to draw. Two residuals remain on the context axis: the released model's absolute entity recall on Earnings-22 drops under the multi-task fine-tune (uplift preserved at +27.5 pp), and its natural-speech distractor hallucination (6.3%) is not covered by the spelled-entity fix. Applying the same counterfactual construction to natural-speech biasing — a follow-up recipe whose context is an entity list that disagrees with the audio, target following the audio — cuts that hallucination to 1.8% (below the base model's 3.7%) while retaining a +25.9 pp uplift, but at a measured cost to absolute entity recall ($+35.3 \rightarrow +24.7$ pp): the natural-speech analog of the spelled-entity result, carrying the recall trade-off the spelled case did not.

The principle, stated once. Two instances now sit side by side (Figure 9). A streaming model may act only on its causal prefix, so its supervision must make the target a function of that prefix and of signals that remain trustworthy at decision time. The temporal leak (labels that peek at the future) manufactured oscillation and a phantom frontier; the contextual leak (a side input that always matched the audio in training) manufactured copying. Same disease — the target depends on something the model cannot trust when it must act — and same cure: a minimal-pair or counterfactual construction that makes the untrustworthy signal uninformative by design.

8 The released system in practice

The release is the merged unified checkpoint, the label-spec builder and training code, the dictation and spelled-entity probes, and the replay benchmark.

Offline cost, measured and attributed. The endpoint fine-tune costs +1.1/+2.7 pp WER on LibriSpeech test-clean/test-other (Appendix C, Table 11). The mechanical explanation — marker fires truncating the decode — is falsified by intervention: banning the marker tokens at decode time drops fires to zero and does not move WER. The regression is distributional drift from a narrow-schema fine-tune; mixing plain transcription examples into the pool (the standard mitigation, exercised in the released model) recovers part of it, and Appendix D documents the full attribution, an AdamW weight-decay hazard for recipes that train a few vocabulary rows against a frozen embedding, and one negative result (Muon underperforms

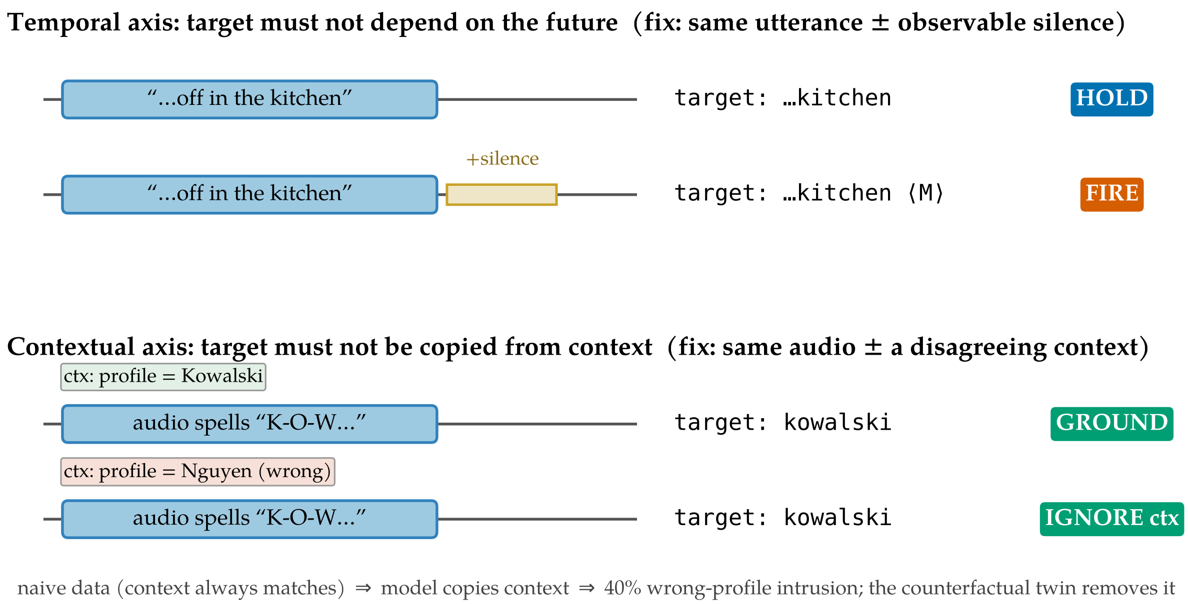

Figure 9. One failure class, two instances. Temporal axis (§4): labels that peek at the future manufacture a phantom frontier; the fix is the minimal pair, the same utterance ± an observable silence tail. Contextual axis (§7): a context prefix that always matches the audio becomes a copyable shortcut; the fix is the counterfactual twin, the same audio with a matching and a disagreeing context, target following the audio.

AdamW on these LoRA factors). Streaming WER, the deployment-relevant number, is unaffected.

Serving on stock vLLM. Three findings from running the model on shared-GPU vLLM [Kwon et al., 2023] rather than a simulator (Figure 10). The natural flat-cost serving shape — appending each 0.5 s chunk as a new audio placeholder in a resident request — is out of distribution for a model trained on single segments and destroys it (88% WER); bounded re-feed of the last window is the correct primitive (3.8%). The vLLM path reproduces the endpoint behavior at 84 ms median per chunk, 5.4× faster than the transformers simulator used during development — latency claims made on $O(T^2)$ re-decode simulators overstate the cost of this approach by that factor. And bounding every resident state matters: a 20 s force-flush fixes both the WER tail and the compute tail, and per-frame P95 stays flat through 8 concurrent sessions on a 0.15-GPU slice (a contention-caveated lower bound; the shape, not the constant, is the result). The in-engine efficiency variant — windowed KV with the `<CTX>` prefix pinned as an attention sink [Xiao et al., 2024] — is validated in a companion systems paper [Li, 2026].

9 Discussion

A checklist. The failure mode generalizes to any model that must act mid-stream on offline-labeled data. Four questions for a streaming supervision pipeline:

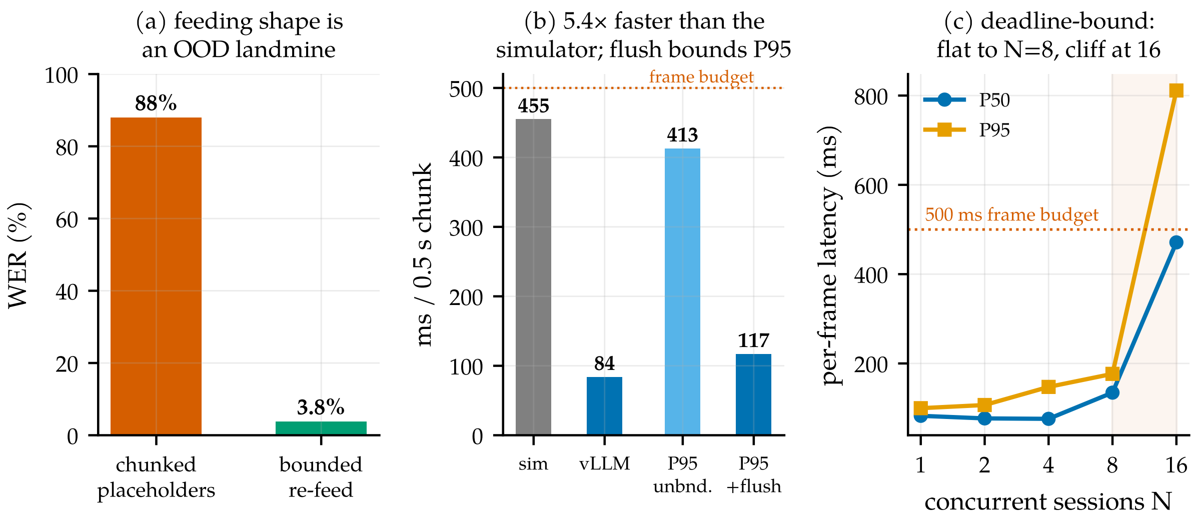

Figure 10. Serving on real infrastructure. (a) Feeding shape: per-chunk placeholders are out-of-distribution; bounded re-feed is correct. (b) Per-chunk compute on vLLM vs. the transformers simulator; the force-flush bounds the P95 tail. (c) Concurrent sessions per engine on a 0.15-GPU slice: flat to N=8, cliff at 16.

Is your supervision clairvoyant?

1. **Prefix test.** For each label at decision time t: is it computable from input up to t? *(Our pause labels were keyed on the next speaker.)*
2. **Twin test.** Can two causally identical prefixes receive different labels anywhere in the data? *(Same complete-utterance prefix: fire in one pool, hold in the other.)* Minimal pairs differing in an *observable* are the fix, not the bug.
3. **Phantom-condition test.** Does the eval demand behavior on conditions deployment never produces? *(Clips cut at end-of-speech; deployment always delivers the next silence.)*
4. **Oscillation test.** Does training bounce between mode attractors that each satisfy part of the data? That is the loss surface reporting a contradiction.

Who else is likely affected. Candidates: turn-taking policies for full-duplex agents trained from offline dialogue transcripts, whose labels encode who *did* speak next rather than what was knowable (the projection objectives of Ekstedt and Skantze, 2022 are explicitly predictive and thus exempt); barge-in and endpointing in commercial stacks; simultaneous translation trained with reference-aligned read/write decisions [Ma et al., 2019]; streaming TTS interruption; and tool-call timing for agents cloned from completed trajectories — a streaming sibling of causal confusion in imitation learning [de Haan et al., 2019]. In each, an apparent aggressiveness-versus-caution frontier may partly be an artifact of asking the model to know the future. The diagnosis joins a lineage of measurement-induced phenomena: as sharp metric choices can manufacture "emergence" [Schaeffer et al., 2023], non-causal label choices can manufacture trade-offs. The mechanism reproduces outside speech: a small transformer on a synthetic stream, with a tunable fraction f of its labels made clairvoyant by construction, trains monotonically at $f=0$ and, as f grows, develops first the fire-versus-hold anti-correlation of a forming phantom frontier and then — once the contradictory pool is large enough to destabilize the single-mode optimum ($f \gtrsim 0.85$) — the same bistable mode-oscillation.

Limitations. Single-speaker, single-channel English; the mixed-channel meeting shape is future work. The dictation capability transfers zero-shot to shorter enumerations (street addresses: 1.00 on-time recall over 60 held-out items) but not to longer-than-trained ones: on 16-digit card numbers the model fires at its learned ∼10-digit completeness point, ∼3 s early, at zero recall, with transcription accuracy unaffected throughout — a learned completeness judge remains the general lever beyond pattern-specific schemas. Folding four behaviors into one rank-16 adapter costs endpointing precision (1.03 versus 0.3 false fires per speech-minute); the pure-endpointing checkpoint remains available for turn-taking-only deployments, and each capability layered on top of it — a larger plain-ASR-replay fraction to recover offline WER, natural-speech biasing counterfactuals to suppress distractor hallucination — buys its target metric at a further, measured cost to that precision. Concurrency numbers are contention-caveated lower bounds from a shared GPU.

10 Related work

Endpointing and end-of-turn detection. Classical endpointing thresholds a VAD [Silero Team, 2021] with a silence timeout, optimized since Raux and Eskenazi [2008]; learned refinements predict end-of-query from acoustics [Shannon et al., 2017], jointly train an endpointer with the recognizer [Chang et al., 2019, Li et al., 2020, Bijwadia et al., 2023], or distill tiny acoustic end-of-turn models [Helwani et al., 2026, Pipecat team, 2025]. Recent systems fuse acoustic and linguistic cues [Wang et al., 2026, Yang et al., 2026] or fold VAD, turn detection, and ASR into one audio-LLM front-end [Li et al., 2026], the latter architecturally closest to us but with no released weights or recipe. Deployed open detectors run beside the recognizer: Smart Turn (acoustic, fully open recipe, no transcript; Pipecat team, 2025), LiveKit's turn detector and TEN (end-of-utterance classifiers over a separate STT's output; LiveKit, 2025, TEN Team, 2025, LiveKit, 2026). Open-weights streaming recognizers with in-model turn signals exist without training accounts: Kyutai STT's semantic-VAD head [Kyutai Labs, 2025] and NVIDIA's Parakeet-EOU token [NVIDIA, 2026] publish weights but neither the labeling procedure nor the data construction, and neither measures context biasing. Commercially, Deepgram Flux [Deepgram, 2026], OpenAI's semantic VAD [OpenAI, 2025], and AssemblyAI's intelligent endpointing [AssemblyAI, 2025] publish no training details. Our contribution to this line is the missing artifact — the recipe — plus evidence that recall-versus-precision trade-offs this literature reports as intrinsic can be manufactured by non-causal supervision.

Turn-taking in dialogue. Turn-taking prediction has a long tradition [Skantze, 2021], from text-based end-of-turn prediction [Ekstedt and Skantze, 2020] to voice-activity projection trained on future windows [Ekstedt and Skantze, 2022]; the human timing evidence comes from Stivers et al. [2009] and Levinson and Torreira [2015]. Projection objectives predict distributions over the future and are causally honest; our diagnosis concerns the sharper failure where future-dependent quantities are presented as deterministic labels.

Streaming and unified ASR. Streaming recognition spans RNN-T [Graves, 2012], large weakly supervised models [Radford et al., 2023], decoder-only streaming with latency losses [Wan et al., 2026], and unified streaming/offline speech-LLMs [Shi et al., 2026]. Kyutai's delayed-streams models add a semantic-VAD head to streaming STT [Kyutai Labs, 2025, Défossez et al., 2024]; full-duplex speech-to-speech models [Défossez et al., 2024] dissolve the turn abstraction entirely and are a different product class. We inherit the unified-ASR substrate and add the in-transcript turn event plus the training recipe.

Context biasing. Contextual ASR ranges from attention-based deep context [Pundak et al., 2018] to retrieval-scale biasing [Gong et al., 2025], RL-tuned hotword integration [Kong et al., 2025], and cue-based prompting for speech LLMs [Novitasari et al., 2026]; Contextual Earnings-22 [Durmus et al., 2026] concurrently standardizes entity-level evaluation on the corpus we use. We use plain prompt-slot biasing and contribute measurements on the open unified-model class.

Serving. vLLM's paged KV [Kwon et al., 2023] carries our serving path; attention sinks [Xiao et al., 2024] and bounded-state streaming serving [Li, 2026] provide the in-engine efficiency variant of our pinned context prefix.

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A Label-specification details

Completeness classifier for pair examples (Table 4, #4–5): utterance A is *incomplete* iff its transcript ends in a filler/connective ("and, but, so, or, uh, um, the, a, to, of, in, with, i mean, you know, ..."), ends mid-word (AMI partial-word annotation), or has fewer than 3 words outside a closed acknowledgment list. Borderline cases land in the `resume_after_fire` cost knob — they never create fire/hold contradictions on identical observables. Abandonment (incomplete speech, speaker never resumes) is an inference-policy timeout, not a label. Pair examples use both same- and different-speaker AMI pairs: the fire decision keys on completeness + silence, never on speaker identity, which single-channel deployment cannot observe.

B Benchmark specification and the corrected classifier

Full protocol: 0.5 s chunks; committed-prefix incremental decoding (emitted text committed, last 5 tokens rolled back) matching the base model's official streaming semantics; fires timestamped at the chunk boundary where <END_SPEECH> first appears; recall window $[-0.25, +1.5]$ s clipped at the next utterance's onset. The first version of the boundary classifier promoted a boundary to turn-final when another meeting speaker interleaved before the target speaker resumed. On a single-channel stream that speech is silence — demanding a fire there is exactly the clairvoyance the benchmark exists to remove. The corrected classification moved the mixed-pools model's recall 0.917→0.906 and the opposed-pools model's 0.935→0.938; no conclusion changed, but the episode is a working example of the §9 checklist applied to one's own evaluation.

C Additional results

Gated arms and the confirm-horizon sweep. Table 7. The confirm horizon behaves as a clean latency-vs-false-fire dial: h=1 strictly improves the mixed-pools model (recall *rises* because deferring the fire lets the segment gather the boundary inside tolerance) and zeroes the opposed-pools and causal models' false fires; h=2 over-suppresses (mixed-pools recall 0.812, latency past budget). The causal model is the only checkpoint whose h=0 row is already deployable. Ungated arms are omitted as uninterpretable (Figure 6): with ∼1.9 spurious fires per second on silence, the pre-diagnosis models score 0.95–0.99 "recall" by chance.

Table 7. All gated arms on the 25-stretch development benchmark. Resume = `resume_after_fire` rate over continuation boundaries (a cost knob, not an error rate).

Arm	Recall	P50	P95	False/min	Resume	$\mathrm{WER}_{\mathrm{mean}}$
offline-labeled + gate	0.896	0.05 s	0.25 s	88.8	0.89	0.85
mixed-pools + gate	0.906	0.16 s	0.89 s	27.3	0.96	0.36
mixed-pools + gate + confirm h=1	0.927	0.68 s	1.27 s	3.2	0.85	0.36
mixed-pools + gate + confirm h=2	0.812	1.15 s	1.42 s	1.1	0.56	0.36
opposed-pools + gate	0.938	0.26 s	0.56 s	3.6	0.93	4.96
opposed-pools + gate + confirm h=1	0.938	0.77 s	1.06 s	0.0	0.85	4.96
causal + gate	0.969	0.39 s	0.70 s	0.3	1.00	1.43
causal + gate + confirm h=1	0.969	0.89 s	1.20 s	0.0	0.93	1.43

The full timeout family. Table 8, both detectors on the development (25 stretches) and fresh confirmation (50 stretches) sets. The family's shape is stable across sets: recall collapses between X=1.0 and 1.5 s (the fire lands outside the +1.5 s tolerance), and no setting reaches the causal model's (recall, latency, false-fire) point. Silero [Silero Team, 2021] ran per-chunk with state resets — a slight handicap; RMS shares the LM arms' exact detector and is the apples-to-apples policy comparison (on this close-talk channel the Silero cascade is dominated by plain RMS).

Table 8. Silence-timeout cascade, all settings, on the development (dev) and fresh confirmation (conf) stretch sets.

Detector	Set	X (s)	Recall	P50	P95	False/min	Resume
RMS	dev	0.5	0.958	0.65 s	0.94 s	6.3	1.00
RMS	dev	1.0	0.969	1.15 s	1.44 s	1.4	0.81
RMS	dev	1.5	0.208	1.38 s	1.49 s	0.8	0.07
RMS	dev	2.0	0.031	0.55 s	0.55 s	0.2	0.00
RMS	conf	0.5	0.978	0.70 s	0.95 s	4.6	0.99
RMS	conf	1.0	0.989	1.20 s	1.45 s	0.6	0.73
RMS	conf	1.5	0.185	1.44 s	1.50 s	0.4	0.07
RMS	conf	2.0	0.016	1.14 s	1.14 s	0.2	0.00
Silero	dev	0.5	0.844	0.54 s	0.88 s	13.5	0.96
Silero	dev	1.0	0.854	1.04 s	1.38 s	5.5	0.85
Silero	dev	1.5	0.417	1.24 s	1.49 s	2.5	0.41
Silero	dev	2.0	0.125	1.32 s	1.42 s	1.0	0.00
Silero	conf	0.5	0.837	0.51 s	0.83 s	12.3	0.88
Silero	conf	1.0	0.864	1.00 s	1.33 s	4.4	0.73
Silero	conf	1.5	0.451	1.29 s	1.48 s	2.0	0.27
Silero	conf	2.0	0.130	1.34 s	1.44 s	1.3	0.07

Fresh-set confirmation. The causal model with the gate on the confirmation set: recall 0.924, P50 0.42 s, P95 0.70 s, 0.21 false fires/min, resume 0.91, WER median 0.30 / mean 4.63 (three long-monologue stretches enter the no-flush repetition loop; the §8 force-flush is the documented fix). The 4.5 pp recall drop from the development set is within the overlap of the two sets' 95% Wilson intervals.

Probe confidence intervals and the dictation scoring window. On the enlarged probes, the released model's 95% intervals are: premature fires per sequence 0.16 [0.12, 0.21], final recall 0.84 [0.79, 0.88], email-with-context 0.94 [0.88, 0.97], intrusion 0.0 [$\leq$0.016]. The shipped scoring window books any fire within +0.25 s of the last speech end as premature and denies it final-recall credit. Re-scoring every fire by its offset δ from the true end of speech (genuine mid-sequence premature: $\delta < -0.30$ s; on-time: $-0.30 \leq \delta \leq +1.75$ s; late: $\delta > +1.75$ s) gives, on the same 250 sequences: 0.008 genuine mid-sequence fires per number (2/250 sequences), 0.992 on-time final recall, zero late fires, and end-fire latency P50 0.40 s (fires quantized up to the 0.5 s chunk grid). The residual under the shipped window is therefore eagerness within one chunk of the true end, not lateness; we report the shipped window in Table 6 for comparability across rows and note that it is conservative for a low-latency endpointer.

Seed robustness of the released recipe. Table 9. Retraining the released recipe (unified pool, weight-decay and cosine fixes, plain-ASR replay) under three seeds and selecting each best checkpoint on the probe axes gives stable premature-fire, context, and intrusion numbers — the unified capability is a property of the recipe, not a lucky checkpoint.

Table 9. Released recipe under three training seeds (best checkpoint each, common probe set). Seed 0 is the released checkpoint.

Seed	Premature/seq ↓	Email +ctx ↑	Intrusion ↓
0 (released)	0.22	0.88	0.000
1	0.14	0.95	0.013
2	0.16	0.88	0.000
mean ± sd	0.17 ± 0.03	0.90 ± 0.04	0.004 ± 0.006

Schema ablation. Table 10. We retrain the endpointing recipe on the v9 pool with one construction removed at a time (iso-count: the dropped quota is redistributed across the remaining schemas, holding volume fixed), select each arm on the holdout composite, and score on the 25-stretch replay set. Each construction has a distinct predicted failure; two of three reproduce it. Removing the **minimal pair** (schema 2) makes silence optional again: gated false fires rise 1.08→5.28 per speech-minute and the confirm horizon cancels 34 candidates versus 10 — the row-4 pathology and its crutch return. Removing the **silence schemas** (7–8) costs silence-competence: ungated, the model emits 1458 markers on silence versus 8, masked in the gated numbers by the energy gate. Removing the **pause pair** (schema 5) does not raise false fires; the model is mildly more conservative (recall 0.948→0.906, false fires 1.08→0.43), so pause discrimination is largely covered by schema 4's complete-A gaps.

Table 10. Schema ablation on the v9 endpointing pool (iso-count; 25-stretch replay, best checkpoint each). "Silence-fires" is the ungated marker count on silence; "h=1 cancels" is the confirm-horizon crutch metric.

Arm	Recall ↑	False/min ↓	Silence-fires ↓	h=1 cancels	Predicted failure?
baseline (full v9)	0.948	1.08	8	10	—
−minimal pair (#2)	0.906	5.28	5	34	yes (silence optional)
−silence (#7–8)	0.969	1.94	1458	12	yes (silence spray)
−pause pair (#5)	0.906	0.43	8	4	no (redundant)

Offline WER regression. Table 11. The fine-tune costs +1.1/+2.7 pp on the full official LibriSpeech splits (single-shot decode on vLLM, jiwer/whisper normalization). "Fires" is the fraction of clips where the model emitted the endpoint marker; offline clips end at or near end-of-speech, and the base model never fires because its marker rows are untrained. Banning the markers at decode time zeroes the fires without moving WER, which rules out decode truncation as the cause; Appendix D attributes the regression to narrow-schema drift and quantifies the plain-ASR-replay mitigation. The released unified model scores 3.7%/7.1%; raising the plain-ASR-replay fraction from 14.5% to 23% shrinks the regression further to

3.5%/6.7%, at a cost to endpointing precision (premature-fire rate 0.16→0.26) — the same fixed-capacity trade-off.

Table 11. Offline WER (LibriSpeech, full official splits).

	WER↓		Fires	
Model	clean	other	clean	other
Qwen3-ASR-0.6B (base)	2.79%	5.14%	0%	0%
+ endpoint LoRA (merged, causal)	3.93%	7.87%	4.7%	5.9%
+ markers banned at decode	3.94%	7.91%	0%	0%

D Training-recipe notes

The unified model (§7) uses the same LoRA fine-tune as the rest of the paper, plus two AdamW hygiene fixes and one negative result. All three were isolated on identical data (the unified pool), so the comparisons below vary only the optimizer setting.

A weight-decay bug worth fixing, but not the WER culprit. We reserve two vocabulary rows as markers and train them by unfreezing `embed_tokens/lm_head` and masking gradients to those two rows. But AdamW's *decoupled* weight decay applies $p \leftarrow p - \eta\lambda p$ to *every* row regardless of gradient, so at $\eta=2\times10^{-4}$, $\lambda=0.01$ the entire (tied) 151k-row matrix shrinks ∼2.4% over 12k steps — silently degrading the base model's vocabulary and output head. This is a general hazard for any recipe that adds a few trainable vocabulary rows to a frozen embedding, and we fix it with a $\lambda=0$ group. The negative half: in a controlled comparison on identical data, the fix (plus cosine schedule) did *not* lower offline WER — the fixed unified checkpoint scores 3.9%/7.3% (clean/other) versus 3.6%/7.0% without it, within checkpoint-selection noise and in the wrong direction. The offline-WER regression is therefore *not* caused by the weight-decay bug; it is narrow-schema drift from fine-tuning on ∼12k examples with no plain-ASR replay data. We keep the fix for its base-model-preservation rationale, not for a WER win we cannot demonstrate.

Plain-ASR replay: the fix that does move WER. The released model mixes ∼2.5k plain transcription examples (audio → transcript, no marker, no context) into the ∼15k-example pool. It recovers offline WER from 3.9%/7.3% (recipe fixes, no replay) to 3.7%/7.1%, and cuts the fraction of offline clips on which the model spuriously fires a marker from ∼10–20% to 1.8%, because the plain-ASR examples teach the model that a complete utterance with no context is simply to be transcribed. A larger replay fraction is the obvious further lever on the residual regression.

Cosine decay. A cosine schedule from 2×10^{-4} to 10^{-5} after a 100-step warmup gives the most stable, highest holdout-composite trajectory (plateau 0.992 from step 6k) and a slightly less trigger-happy final checkpoint than the constant schedule.

Muon underperforms AdamW here. We tried Muon [Jordan et al., 2024] on the 2-D LoRA factors (Newton-Schulz-orthogonalized momentum, AdamW retained for the marker rows) at a $100\times$ larger base LR (2×10^{-2}, since the orthogonalized update is $\sim$unit-norm). On identical data Muon's raw loss descends *faster* early but its task metric plateaus at holdout composite 0.846 (by step 3–4.5k, conversational schema collapsing) versus AdamW's 0.985. The cause is structural: rank-16 LoRA factors (1024×16) are too rectangular for the 5-step Newton-Schulz iteration to orthogonalize (empirically $\|O^\top O - I\|$ off-diagonals $\approx$0.5), and orthogonalizing the two factors separately is not the same operation as orthogonalizing their product. We keep AdamW.

E Reproducibility

Hardware: one RTX Pro 6000 (Blackwell), shared with an unrelated co-tenant workload throughout (all GPU jobs wrapped in retry/resume; the causal training run survived four OOM kills via checkpoint resume with $\leq$750 steps lost each). Training: LoRA r=16, α=32, batch 8, AdamW with the embedding/head weight-decay and cosine-schedule fixes of Appendix D; the unified checkpoint is selected at step 7.5k on the probe axes (dictation, context, intrusion) rather than the holdout composite alone. Evaluation and serving: vLLM 0.19, `skip_special_tokens=False` (the markers are reserved special tokens and are otherwise silently stripped — a gotcha worth one sentence). Code, the label-spec builder, the dictation/spelled probes, the replay benchmark, and the merged unified checkpoint are released at `https://github.com/19PINE-AI/turn-aware-asr`; the repository README documents the mapping from its internal checkpoint directory names to the recipes described in this paper.